

Technical Report

1. Introduction:

Customer churn is one of the most pressing challenges for subscription-based businesses. Losing customers not only reduces revenue but also increases acquisition costs, as replacing lost customers is significantly more expensive than retaining existing ones.

This project analyzes a dataset of **500+ customer records** to identify churn drivers, build predictive models, and recommend actionable retention strategies. The analysis follows a structured workflow:

- Data Cleaning and Preparation
- Exploratory Data Analysis (EDA)
- Advanced Modeling and Segmentation
- Business Recommendations and Implementation Plan

2. Data Preparation:

Dataset Overview:

- **Size:** 500+ rows
- **Key Attributes:**
 - Tenure (months)
 - Monthly Charges
 - Total Charges
 - Contract Type (Month-to-Month, One Year, Two Year)
 - Payment Method (Electronic Check, Credit Card, Bank Transfer)
 - Paperless Billing (Yes/No)
 - Senior Citizen (Yes/No)
 - Churn (Target Variable)

Cleaning Steps:

1. Converted TotalCharges to numeric and handled invalid entries.
2. Removed missing values (~2% of rows).
3. Encoded categorical variables using Label Encoding.
4. Standardized numerical features for modeling.
5. Saved outputs into data/cleaned_data.csv.

3. Exploratory Data Analysis (EDA):

EDA was conducted to uncover churn patterns and correlations.

Visualizations:

1. **Churn Distribution:** ~26% of customers churned.
2. **Monthly Charges vs Churn:** Higher charges correlated with higher churn.
3. **Correlation Heatmap:** Tenure negatively correlated with churn.
4. **Tenure vs Churn Rate:** Customers with shorter tenure churn more frequently.
5. **Contract Type vs Churn:** Month-to-month contracts had the highest churn.

Insights:

- Customers with **short tenure** and **high monthly charges** are most vulnerable.
- **Electronic check payments** are strongly associated with churn.
- **Senior citizens** churn more frequently than younger customers.

4. Advanced Analysis:

4.1 Predictive Modeling

- **Logistic Regression:** Provided interpretable coefficients for churn drivers.
- **Random Forest Classifier:** Achieved higher accuracy and ROC-AUC (~0.85).
- **Evaluation Metrics:** Precision, recall, F1-score, and ROC curve.

Result: Random Forest identified contract type, payment method, and monthly charges as top predictors.

4.2 Customer Segmentation (Clustering)

- Applied **KMeans clustering** on tenure and charges.
- Identified three customer segments:
- **Cluster 1:** Low tenure, high charges → highest churn risk.
- **Cluster 2:** Long tenure, moderate charges → stable customers.
- **Cluster 3:** Mid-tenure, low charges → moderate risk.

4.3 Survival-Style Analysis

- Plotted churn probability across tenure.
- Showed a steep drop in churn after ~12 months, indicating customers who stay longer are less likely to leave.

5. Key Insights:

- **Month-to-month contracts** and **electronic check payments** are strongly associated with churn.
- **Senior citizens** churn more frequently than younger customers.
- **High monthly charges** increase churn risk.
- Tenure is a stabilizing factor: customers who stay longer are less likely to churn.

6. Recommendations:

1. **Contract Incentives:** Offer discounts or loyalty rewards for customers switching to annual contracts.
2. **Payment Optimization:** Promote auto-pay or digital wallet options.
3. **Targeted Support:** Tailored engagement programs for senior citizens.
4. **Pricing Strategy:** Loyalty discounts for high-charge customers.

7. Business Implementation Plan:

Step 1: Deploy churn prediction model to flag high-risk customers.

Step 2: Launch targeted retention campaigns (contract incentives, payment method upgrades).

Step 3: Monitor churn monthly and adjust strategies.

Step 4: Measure success via churn reduction percentage and revenue retention.

Expected Impact: Reduce churn by **10–15%**, translating into significant annual revenue retention.

8. Conclusion:

This project demonstrates how data-driven insights can reduce churn and improve customer retention. The combination of predictive modeling, segmentation, and survival analysis provides a robust framework for proactive customer retention strategies.

9. Appendix:

- **Code Notebooks:**
 - 1_data_cleaning.ipynb
 - 2_eda.ipynb
 - 3_analysis.ipynb
- **Data Files:**
 - data/raw_data.csv
 - data/cleaned_data.csv

- **Reports:**

- executive_summary.docx
- technical_report.docx

- **Presentation:**

- business_presentation.pptx